

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2023

Bachelor in Computer Applications

Course Title: **Distributed System**

Code No: **CACS352** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is middleware? Explain the different architecture styles of distributed systems.
3. Differentiate between stateful and stateless servers.
4. Explain Berkeley algorithm with a suitable diagram.
5. Explain the types of data-centric consistency models.
6. Define distributed commit. Explain two-phase commit with a suitable diagram.
7. Describe security management in the context of distributed systems.
8. Write short notes on (any two): a. Sequential consistency b. Access control matrix c. Message passing interface

Group C

Attempt any TWO questions[2x10=20]

9. What is a distributed system? Explain the characteristics of distributed systems and different types of Distributed Systems.
10. What is an election algorithm? Explain the different election algorithms with a suitable diagram.
11. Explain Remote Procedure Call and its working process with a suitable diagram. Furthermore, provide an explanation of message-oriented communication, along with an illustrative example.